

Reflection

I created a basic prototype in two stages before using AI. Printed appointments from separate variables, then a list of dictionaries using `book_appointment()` and `display_appointments()`, and a check for a blank patient name. It was running, but I identified at least five drawbacks: hard-coded data, free-text times, no double-booking check, no search/cancel, and only the patient name was validated.

I learned from AI why those restrictions are important, particularly that a time stored as plain text can't be compared or sorted, and that checking just one field leaves the others unprotected. It also asked me questions that made me trace what happens with a `None` time.

Yes, AI assumed. Its alternative removed all three fields as required text, added type checking and whitespace trimming, which the client has not requested.

I tested both versions and compared the output on normal, blank, `None` and duplicate-booking inputs, and wrote unit tests (`test_stage01.py`). That demonstrated that the AI version was more strict with the input, but still permitted double booking.

The engineering work that was left was mine: what constitutes a duplicate, what improvement to make, and what fields are actually needed by the client.